

CODEL

Electrical Continuity Test Certificate

Customer : Viet An Environment

Customer Order No. : SAP 4037

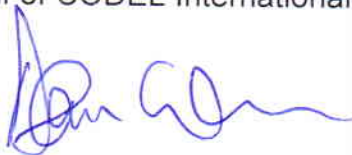
Model No. :

Serial No. : SCU-1981(CEMS)

This is to certify that the electrical components on the above equipment have been tested for electrical continuity, earth leakage and insulation and that all tests have been completed successfully.

Signed on behalf of CODEL International Ltd.

Sign :



Print name : A Whitaker

Date : 20/12/16

CODEL

Pneumatic Leak Test Certificate

Customer : Viet An Environment

Customer Order No. : SAP 4037

Model No. :

Serial No. : SCU-1981(CEMS)

This is to certify that the pneumatic components on the above equipment have been tested for pneumatic leaks and that all tests have been completed successfully.

Signed on behalf of CODEL International Ltd.

Sign :



Print name : A Whitaker

Date : 20/12/16

20/12/16

Order: SAP 4037
 Customer: Viet An Environment
 Site:
 Station: GCU 40 TUV

SCU OUTPUT CONFIGURATION

Current Outputs from 802-014

Response	Source	Address	Type	Range
Analysers	1 NO	#1	4-20mA	0 -> 1000mg/Nm3
Analysers	2 CO2	#1	4-20mA	0 -> 25.0%
Analysers	3 SO2	#1	4-20mA	0 -> 1000mg/Nm3
Analysers	4 H2O	#1	4-20mA	0 -> 25.0%
Analysers	5 DUST	#1	4-20mA	0 -> 500mg/Nm3
Analysers	6 FLOW	#1	4-20mA	0 -> 1000000Nm3/h
Analysers	7 Temp	#1	4-20mA	0 -> 600°C
Analysers	8 Pressure	#1	4-20mA	0 -> 160kPa

Relays Outputs from 802-024

	Source	Address	Function	Level	Response
1	GCEM	#1	Data Valid		Immediate
2	GCEM	#1	Calibration		Immediate
3	GCEM	#1	Purge		Immediate
4	GCEM	#1	Plant Status		Immediate

System Arrangement

SCU- -----#1 GCEM
 -----#2 DCEM
 -----#3 VCEM

2@ 4x4-20mA output card fitted 802-014B
 1@ 8xRelay output card fitted 802-174

EPROMS
 Slave 507-034D
 Master 507-035ALE

Configured by
 A Whitaker

CODEL INTERNATIONAL

MODEL Dcem 2000/2100 Test Certificate

Test Certificate No. 318

	Serial Numbers				Batch No./Works Order No.		Part No./Rev.	
Transceiver	DCEM2000-318TRX1				1/1	4037	990.876N	
Transceiver	DCEM2000-318TRX2				1/1	4037	990.876N	
Signal Processor	DCEM2000-318SPU				1/1	4037	990.871J	
Data Display Unit								
Power Supply								
EPROM Set								
Customer	VIET AN				Order No.	4037		
Location	Vietnam				Date of Test	25/11/2016	Tested by H. M	
Cable	10m	✓	Other		LSF		Pathlength	2m ✓ Other

Test Data

Detector Levels	Levels							
	D t 1 14480		D r 1 20458		T12	100.01%	K12	7129
	D t 2 14475		D r 2 20425		T21	99.97%	K21	7149
Diagnostic and Calibration Values	0mA count (1) : N/A		20mA count (1) : N/A		O/p current (1) :	0-20mA	N/A	4-20mA N/A
	0mA count (2) : N/A		20mA count (2) : N/A		O/p current (2) :	0-20mA	N/A	4-20mA N/A

Test Equipment

Scope Serial No : TDS1002B C104574	Meter Serial No: 53650822	Test Filter Serial No: 4399 (53.5%) 53.6
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Operation Checks

Normalising Inputs	Temperature	Keypad	Serial Comms.	N/A	Plant Status	N/A
	Oxygen	Keypad	Data Valid LED	N/A	CONTACT	N/A
	Pressure	Keypad	Alarm LED	N/A	CONTACT	N/A
	Water Vapour	Keypad	Alarm Level 1/2 %	N/A	Span Check	N/A

Special Details

Date on Soak : 24/11/2016		Date off Soak : 28/11/16	
Pass/Fail : Pass		Signed : H.Medipally	

Test Certificate number :	340	VCEM 5000/5100 :	VCEM5000
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Serial Numbers		Batch No. / Works Order No.		Part No.	Revision
Transducer Rx1	VCEM5000-340 -RX1	1/1	4037	990.483	D
Transducer Rx2	VCEM5000-340 -RX2	1/1	4037	990.483	D
Signal Processor	VCEM5000-340 -SPU	1/1	4037	990.873	F
Data Display Unit					
Power Supply					

Customer	VIET AN Enviromental	Order No.	4037
Location	VIETNAM	Date Tested	22-11-2016
		Tested By	D. M

Flow check

Flow rate over soak period	
Target:	4.0
Actual:	4.0

EPROM Set

SP Slave:	507-030A	DDU:	N/A
SP Master:	507-031A		

Linearity

Simulated Value	Displacement Value		Data Rate	
	Target	Actual	Target	Actual
1m/s	125 ±1	126	8	8
4m/s	125 ±1	126	2	2
10m/s	100 ±1	100	1	1
20m/s	100 ±1	100	0	0
40m/s	50 ±1	50	0	0

DDU Configuration

mA outputs		mA range	Flow range	Relay outputs		Alarm level
	Output 1	4-20mA	0-40m/s		Relay 1	N/A
	Output 2	4-20mA	0-40m/s		Relay 2	N/A

Special Details:

This is to certify that the products listed above have been tested to **CODEL's** standard procedures and that they operate within specifications.

Signed :	D. Morrell	Date:	22-11-2016
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Test Certificate number : 1255

Serial Number	Batch Number / Works Order Number		Part Number/Revision
PSU48-1255	1/1	4037	990.851H

Customer	VIET AN
Location	Vietnam
Order Number	4037

Test Equipment: MEGGER PAT 3

Date Tested : 30/11/2016 By : A. N

	PASS	FAIL
Continuity Test:	✓	
Earth Continuity Test:	✓	

	Polarity check	Voltage	Voltage Under Load
Output 1 :	✓	48.28	47.86
Output 2 :	✓	48.28	47.87
Output 3 :	✓	48.28	47.97
Output 4 :	✓	48.28	47.96

Power Rating : 125W

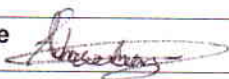
Special Details :

Counter Signed :



This is to certify that the products listed above have been tested to CODEL's standard procedures and that they operate within specifications.

Signed: A. Nnaedozie



Date : 30/11/2016

TEST CERTIFICATE

Reference Gas Data

Gas	Concentration ppm	Gas Supplier	Cylinder No.
SO2	3018	Air Liquide	VC4053
CO2	100%	Air Liquide	VC4337C
NO	1996	Air Liquide	VC6647A

Calibration Data

Channel No	Blended Concentration ppm	Measured Reading ppm	% Error
1(NO)	507.8	507.7	<1%
2(SO2)	255.96	255.21	<1%
4(H2O)	42.2	43.3	<1%
5(CO2)	25.3	25.3	<1%

Operational Checks

Normalising Inputs & Ranges	Temperature °C		600
	Detector Heat sink Set Point °C		52
	Oxygen %		25
	Pressure kPa		160
EEPROM Nos.	Master	507- 067A	Slave 507- 066A

Special Details:

Test Certificate No.: 40series1007

Pass



Fail





TEST CERTIFICATE

Test Certificate No.: 40series-1007

Equipment: GCEM40series-TUV

Serial No.: 1007

Part No./Rev.: 991.470A

Customer: Viet An Environment Technology Co. Ltd.

CODEL Order No.: 4037

Date of Test: 28/11/2016

Tested by: A Aras

Signed:

A handwritten signature in blue ink, appearing to read "A Aras", followed by a stylized flourish or second signature.

on behalf of:

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